

Independent Topic Form (3 credits)

Independent study of a particular topic under the supervision of an instructor with relevant expertise in the area.

This form must be returned to the MMA Office complete with your supervisor's signature by the Add/Drop Deadline (2nd week of classes) of the term in which you plan to undertake this project.

Name: Mustafa Yousuf McGill ID#: 261265412

Course Number: BUSA 613 Term of Project: _____

Proposal Title: Detecting and Explaining Health Misinformation Narratives Using NLP and Social Engagement Analytics

Brief Description:

This study will examine how natural language processing and social media analytics can be used to detect, explain, and monitor health misinformation narratives online. Using publicly available datasets such as CoAID and FakeHealth, the project will compare traditional machine learning models, transformer-based language models, and hybrid models incorporating social engagement signals. Additionally, the study will use topic modeling, sentiment/emotion analysis, and explainability techniques to identify recurring misinformation narratives, evaluate model failure patterns, and assess whether engagement features improve detection. The final output will be a reproducible analytical pipeline and research report with practical implications for public-health communication, platform governance, and digital risk monitoring.

Basis on which grade will be calculated:

A progress report and a final report of no more than 5 pages, along with source code, will be used to grade

Student's Signature: Mustafa Yousuf Date: April 28, 2026

Supervisor's Name: Taha Havakhor, Ph.D.

Supervisor Signature: Havakhor Date: Apr 28, 2026